



WiFi Shuriken & Wardriving History

CoD_Segfault



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The background of the slide is a dark blue gradient. On the left side, there is a complex, abstract pattern of light blue lines that resemble a circuit board or a network diagram. These lines are interconnected and form various geometric shapes. In the top-left corner, there is a small cluster of these lines that form a path leading to a circle containing the letter 'T'. Below this, there is another circle containing the number '1'. Further down, there are two small circles, one containing a plus sign and the other a minus sign, connected by lines. The overall effect is a high-tech, digital aesthetic.

T

1

01

Who Am I?

CoD_Segfault

- WiGLE Rank: 38 (as of 2026-05-01)
- Author of BW16-Open-AT
- Code Contributor to Wardriver.uk
- Hardware Designer
 - Mini Wardriver v1 and v2
 - WiFi Shuriken
- Hard Hat Brigade Co-Organizer





02

What is the WiFi Shuriken?

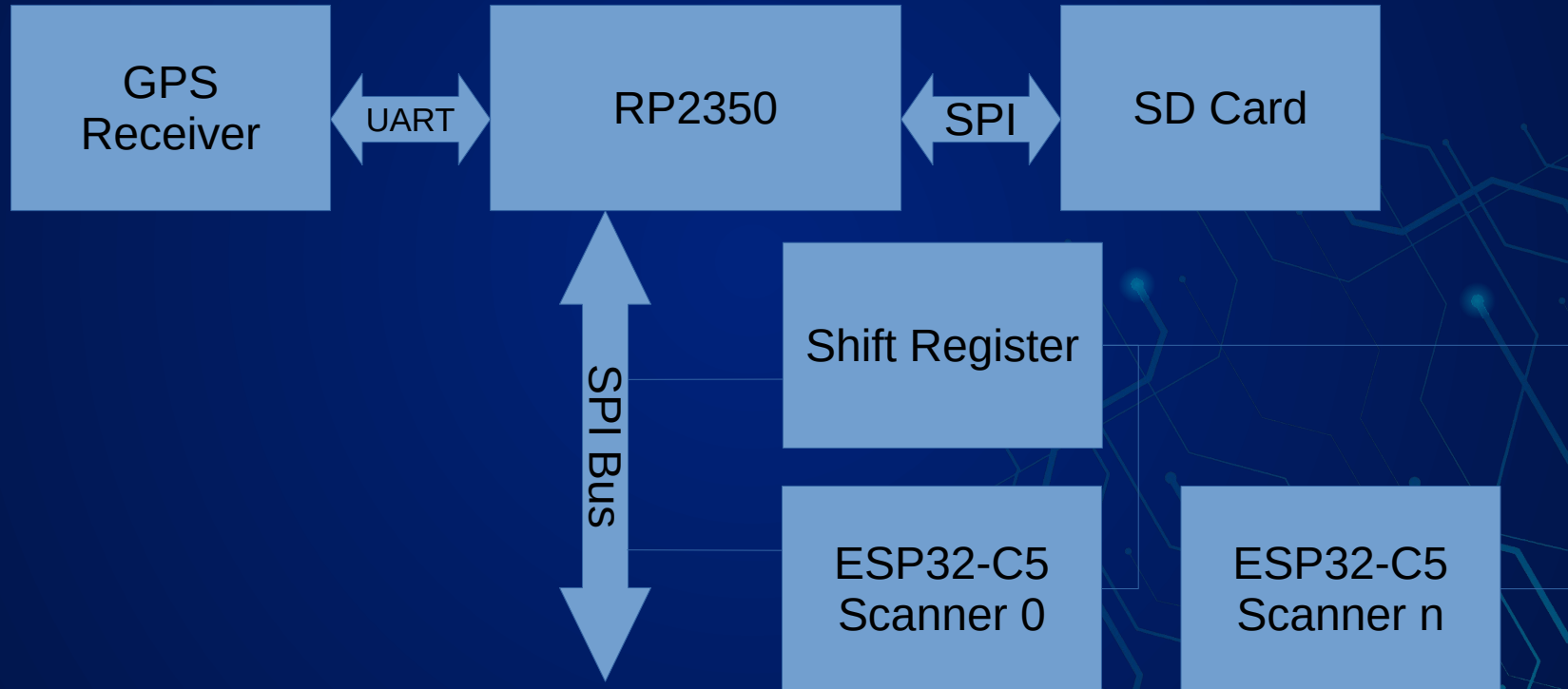


What is the WiFi Shuriken?

- Microcontroller-based Wardriving Platform
- Standalone Operation
 - Six Dual Band WiFi Radios
 - Multi-Constellation GNSS
 - SD Logging
- Controller and Scanner Architecture
- Engineered to Remove Bottlenecks

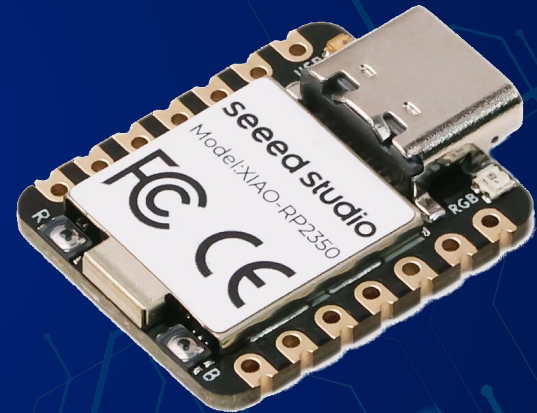


Shuriken Architecture



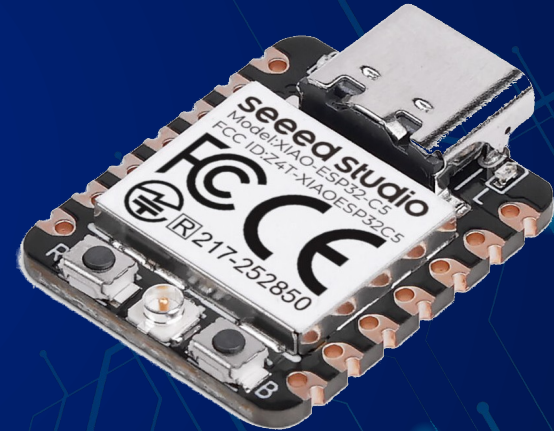
Shuriken Controller

- Seeed Studios XIAO RP2350 (Raspberry Pi Pico 2)
 - Dual-Core ARM Processor
 - Used in DC32 Badge
- Core 0
 - SD Card Logging
 - GPS Location, Time, and Date
 - Optional Phone/PC Integration (In Development)
- Core 1
 - Scan Channel Scheduler
 - Scanner Communication



Shuriken Scanners

- Seeed Studios XIAO ESP32-C5
 - Single-Core RISC-V Processor
 - Dual-Band WiFi Radio
- Accepts Commands From Controller
- Handles All WiFi Scan Logic
 - Radio Control (Active/Passive, Scan Duration)
 - Queues Results Until Queried



Operation

- Apply Power; Scanning and Logging Begins
- Deduplication of Observed Devices
 - SSID – Network Name
 - BSSID – MAC Address
 - Band and Channel
 - Security Set
- Fault Tolerant
 - No Single Scanner is Responsible for Single Channel



Scalable Future



- Potential Future Scanners
 - HaLow
 - 6 GHz
 - ???
- More Scanners
 - Trivial to Add 64+ Scanners



03

What is Wardriving?

- Observation of Wireless Devices

- WiFi Access Points
- Bluetooth Devices
- Cellular Networks

- Metadata Collection

- SSID & BSSID
- Band and Channel
- Security Set
- Signal Strength
- Location
- Date and Time

What is Wardriving?

Wardriver

SecureThisNow

\ wôr'drīvər \ *noun*

person who spends an
unreasonable amount of
money and time on the hobby
of searching out and mapping
wireless networks

Also see: stumbler, wiggler

Image Credit: MrBill

What To Do With the Data?

- WiGLE – Wireless Geographic Logging Engine
 - User-Submitted Submissions
 - Catalogs and Indexes Observed Networks
 - Trilaterates to Determine Location
 - Displays Results on a Map
 - Generates Stats with Trends Over Time



Why?

- Fake Internet Points!
- Collectively See Trends
- Explore New Areas
- It's Fun

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04

**How Did We
Get Here?**

The Beginning - PC

- Laptop
 - Netstumbler (Windows)
 - Kismet (Linux)
 - KisMAC (Mac OS X)
- WiFi Card
 - PCMCIA, USB, Internal
- USB GPS



Somewhere Along the Way - PC

- Dedicated Builds
 - Networked Clusters
 - Dozens of WiFi Cards
 - Antenna Distribution Systems
 - Power Distribution

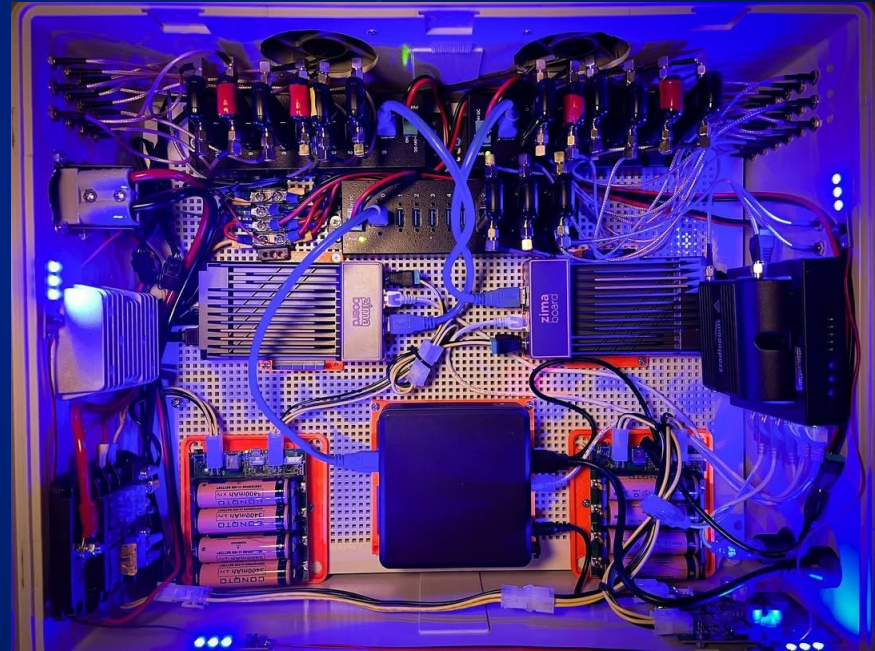


Image Credit: BusySignal

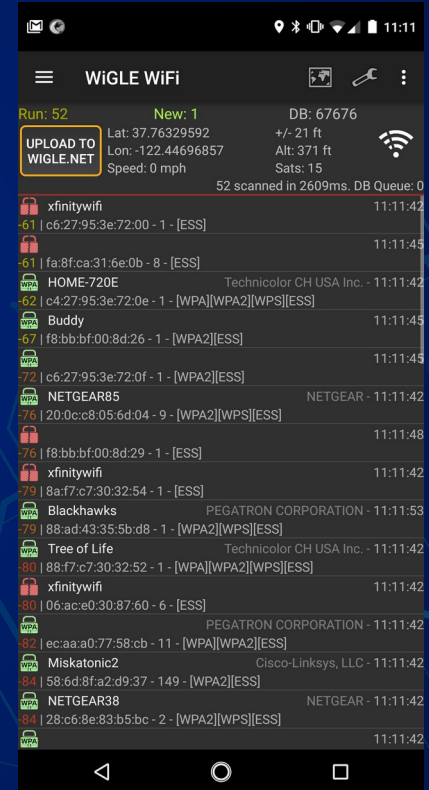
ARM Single Board Computers

- Raspberry Pi
 - Lower Power Requirements
 - Full Linux Support
 - Compatible with Kismet
- USB Expansion



Android Phones

- WiGLE App
 - Free
 - Available From Play Store or GitHub
 - Automate via Tasker
- Always With You



WiFi Microcontrollers

- Espressif
 - ESP8266
 - ESP32
- Realtek
 - RTL8720DN



- Joseph Hewitt Made the First Version in 2019.
 - Arduino with an ESP8266 Running AT Firmware.
- Version 2 Was 2x ESP8266 chips.
- Version 3 Upgraded to 2x ESP32 Chips.
 - Added Bluetooth, GSM.
 - Later Replaced GSM with BW16.

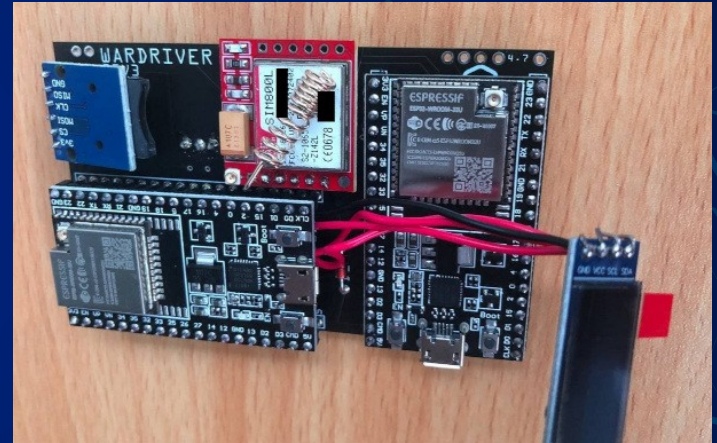
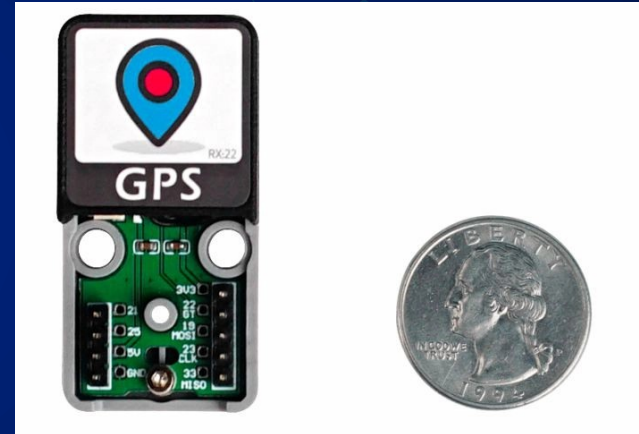


Image Credit: Joseph Hewitt

M5 Atom GPS

- Originally implemented by Lozaning
- Major code contributions by Luke Switzer
- Ultra compact



WiFydra

- Designed by Lozaning
- Up to 16x ESP32-C3 Or ESP32-S3 set to each scan individual channels.
- Covers Entire 2.4 GHz WiFi Band Simultaneously

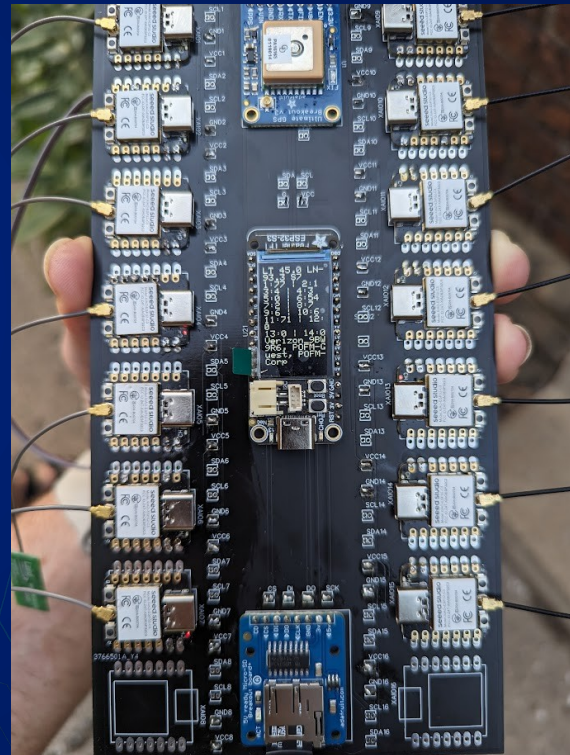
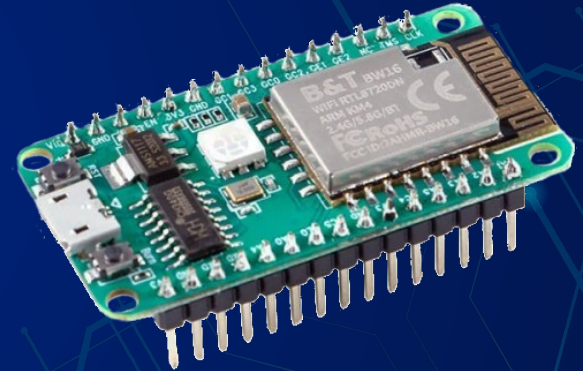


Image Credit: Lozaning

Realtek RTL8720DN

- First 5GHz Microcontroller
- Wardriver.uk Adapted to Use B&T AT Firmware
- Newer Revisions Shipped with Incompatible Firmware
- BW16-Open-AT
 - Open Replacement for the B&T firmware.
 - Subset of AT Commands for Wardriver.uk
 - Properly Handles Modern Encryption



Espressif ESP32-C5

- First 5GHz in ESP32 Line
- Mostly Code Compatible with Previous ESP32s
- More Flexible Radio than Realtek



What Makes the Shuriken Unique?

- Extreme Separation of Responsibilities
- Intentionally Limited Scope
- Fast, Direct Communication
- Fault-Tolerant
- Scalable
- Future-Proof





05

Q&A



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Thanks!

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